Aditi Singhania

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PROFESSIONAL SUMMARY

Results-driven software engineer with strong foundation in distributed systems, high-concurrency backend services, and clean architecture. Experienced i

EDUCATION

B.E. in Computer Science — International Institute of Information Technology *2021 - 2025*

Cumulative Grade Point Average: 8.30/10.0

WORK EXPERIENCE

Software Engineer - Core Services — Zepto (KiranaKart Technologies), Mumbai, India *Jun 2024 - Present*

- Optimized database queries and added composite b-tree indexes on partitioned ledger tables, lowering average query execution time from 420ms to 3
- Architected high-throughput payment webhook processor using Go and Redis Streams, handling 14,000 requests/sec with p99 latency under 28ms.
- Implemented automated circuit-breaker pattern with Resilience4j across 8 downstream microservices, preventing cascade outages during third-party g

TECHNICAL PROJECTS

Peer-to-Peer File Transfer Protocol with NAT Traversal | Technologies: Rust, STUN/TURN, UDP, Tokio, Cryptography

- Developed user-space P2P file synchronization tool implementing UDP hole-punching and ICE protocol to bypass restrictive symmetric NATs.
- Utilized ChaCha20-Poly1305 authenticated encryption to guarantee confidentiality and packet integrity over public Wi-Fi networks.
- Achieved wire-speed throughput of 940 Mbps over local gigabit connections using zero-copy memory-mapped file buffers.

Distributed Log-Structured Key-Value Store | Technologies: Go, Raft Consensus, gRPC, LSM-Tree, Docker

- Implemented distributed LSM-tree storage engine in Go with write-ahead logging (WAL), SSTable compaction, and bloom filters.
- Embedded Raft consensus algorithm for distributed multi-node replication, surviving arbitrary node crashes without state divergence.
- Benchmarked performance using custom concurrent load generator, achieving 85,000 read operations/sec across a 5-node cluster.

TECHNICAL SKILLS

Languages: Go, Java, Python, TypeScript, C++, SQL

Frameworks & Systems: Spring Boot, Node.js, FastAPI, React, Next.js, gRPC

Databases & Storage: PostgreSQL, Redis, MongoDB, Apache Kafka, Elasticsearch

Cloud & DevOps: Docker, Kubernetes, AWS (EC2, S3, RDS), Git, CI/CD Pipelines